

Pocket PM^{v9}

System Architecture

Bid Bigger. Build Smarter. Win More.

Browser → Railway API → PocketBase VPS → Anthropic Claude → External Integrations

Page 2	Browser pocket_pm_v9.html 27 modules vanilla JS no build step
Page 3	Railway API Express.js Node 20 auth proxy Claude proxy 11 AI endpoints
Page 4	PocketBase VPS Ubuntu 22.04 + Caddy SQLite 21 collections file storage
Page 5	Anthropic Claude claude-sonnet-4 domain-tuned prompts RS Means methodology
Page 6	External Procore export ACC export RS Means Phase 2 roadmap

App Core

Single HTML file — no build step, no bundler
27 static panels shown/hidden by nav()
Single `<script>` block — strict no-backtick rule
Offline demo data fallback via localStorage
Python validation after every build

Auth & Session

Login POSTs to Railway `/api/auth/login`
JWT token stored in localStorage as `ppm_token`
Bearer header sent on every API call
Session restored on page reload automatically
`ppmLogout()` clears token and reloads

API Client Layer

`_rail(method, path, body)` — all Railway calls
`apiList / apiCreate / apiUpdate` — CRUD helpers
`callAI(messages, system)` — Claude via Railway
`USE_RAILWAY` flag — true when real URL is set
Direct Anthropic fallback for dev/demo mode

HTTPS | JWT Bearer token | JSON -> Railway API

27 Modules: Dashboard | Submittal Registry | Prequal Builder | Drawings | Submittals | RFIs | Punch List | Budget & Cost | CPM Schedule | DFO Manager | 3-Phase Inspection | Deficiency Tracker | PDCA Dashboard | AI Estimating | Change Orders | Pay Application (G702/703) | AIA Contract Dashboard | AIA Document Library | Clause Risk Scanner | Risk Register | AIA Notices | AIA Subcontracts | AIA Closeout | OSHA Safety Management | Safety Plans | PM Assistant | Daily Log AI

Coding Rules: No backtick template literals | No curly/smart quotes in JS | Single `<script>` block only | No inline onclick with dynamic IDs | All modal IDs globally unique | Reserved: `_liveSubs2` for Subcontracts module | Each module: `load[X]Live()` → `_render[X]()` → `wire[X]Panel()`

Demo Project: Riverside Medical Building C — \$4.2M AIA A101 commercial healthcare — Charlotte, NC — Day 142 of 245

Auth Middleware

requireAuth() wraps every protected route
Validates Bearer token against PocketBase
401 returned before any business logic runs
requireRole() for principal / pm / superintendent
Per-request userPb() for row-level PB rules

Claude AI Proxy

ANTHROPIC_API_KEY lives in Railway env only
Key never leaves the server — never in HTML
POST /api/claude — general AI assistant
POST /api/claude/rfi-draft — AIA G716 format
POST /api/claude/co-pricing — RS Means method
POST /api/claude/safety-analysis — 29 CFR 1926
POST /api/claude/contract-review — AIA A201
POST /api/claude/toolbox-talk — OSHA format
POST /api/claude/daily-log — construction report
POST /api/claude/estimate — returns JSON array
POST /api/claude/notice-draft — AIA notice letter

CRUD + Sync Routes

makeCrudRouter() factory — GET/POST/PATCH/DELETE
Project-scoped filters enforced server-side
/api/projects — with dashboard aggregate stats
/api/submittals, /api/rfis, /api/dfows
/api/deficiencies, /api/change-orders
/api/daily-logs, /api/notices, /api/estimates
POST /api/sync/procore-export/:id
POST /api/sync/acc-export/:id

PocketBase JS SDK | Admin service account | Internal network -> PocketBase VPS

Full Route Map: GET|POST /api/auth/login | POST /api/auth/refresh | GET /api/auth/me | GET /api/projects | GET /api/projects/:id | GET /api/projects/:id/dashboard | POST /api/projects | PATCH /api/projects/:id | POST /api/claude (x8 specialized) | /api/submittals | /api/rfis | /api/dfows | /api/deficiencies | /api/change-orders | /api/daily-logs | /api/notices | /api/estimates | /api/sync/procore | /api/sync/acc | GET /health

Security Stack: Helmet security headers | CORS: allowedOrigins[] whitelist (app.pocketpm.fyi) | express-rate-limit: 200 req/15 min global | Claude limiter: 20 AI calls/min | compression middleware |morgan request logging | JWT verify on every protected route

PocketBase Client: Singleton admin client — authenticates on startup | Auto-refresh token every 55 min | pbList / pbGet / pbCreate / pbUpdate / pbDelete helpers | userPb(token) factory for per-request user context

21 Collections

migrate.js seeds all 21 collections on setup
Node.js script — run once on first deployment
All records scoped via project relation field
Cascade deletes remove child records on project delete
Select fields for status / severity / type enums

File Storage

Drawings (PDF) — up to 100 MB per file
Daily log photos — up to 10 MB per photo
Submittal attachments — up to 50 MB
Deficiency photos — up to 10 MB
Stored in pb_data/storage/ on VPS disk

Infrastructure

PocketBase Go binary — internal :8090 only
Caddy reverse proxy — HTTPS with CORS headers
Let's Encrypt SSL — auto-renews every 90 days
UFW firewall — ports 22, 80, 443 only
pb_public/index.html serves the frontend app

All 21 Collections: projects | users | subcontractors | submittals | rfis | dfow | deficiencies | daily_logs | change_orders | pay_applications | schedule_items | punch_list | drawings | aia_notices | budget_items | ai_sessions | safety_observations | invitations | tasks | closeout_items | contract_notices

Caddyfile: app.pocketpm.fyi -> /var/www/pocketpm (static files) | pb.pocketpm.fyi -> 127.0.0.1:8090 (PocketBase) | api.pocketpm.fyi -> 127.0.0.1:3000 (Express) | CORS: Access-Control-Allow-Origin * on all origins | OPTIONS -> 204 (preflight handled at Caddy level)

Cost (Production): Hetzner CX21 VPS: ~\$6/mo (2 vCPU / 4 GB RAM / 40 GB SSD) | Domain: ~\$1/mo | SSL: Free (Let's Encrypt) | Total infra: ~\$7/mo | At 100 subscribers x \$149/mo = \$14,900 MRR — infra is 0.05% of revenue

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Model & Context

Model: claude-sonnet-4-20250514
Max tokens: 4096 per call (capped in Railway)
Rate limit: 20 AI calls per minute per IP
Live project context injected into every prompt:

- Project name, contract value, city/state
- Open RFIs, deficiencies, active DFOWs
- Schedule status, subcontractor issues

Domain Prompts

AIA G716 RFI format — design clarification letters
AIA G701 Change Order — RS Means pricing method
AIA G702/G703 Pay Application — SOV format
29 CFR 1926 OSHA — citation-level safety analysis
AIA A201 Article-by-article contract risk scanner
CQM-C quality patterns — deficiency classification
RS Means methodology — CCI factors by city/state
CSI MasterFormat — division structure for estimates

Structured Outputs

Estimate endpoint returns pure JSON array
Each item: div, description, unit, qty, costs
Markdown fences stripped before JSON.parse()
All other endpoints return plain text strings
PM Assistant maintains AIHIST[] multi-turn history
AI sessions optionally saved to ai_sessions collection

On-demand export. | No Procure or ACC subscription required -> External Integrations

AI Endpoints (11): POST /api/claude — general PM assistant with full conversation history | /rfi-draft — AIA G716 format | /co-pricing — RS Means methodology with CCI | /safety-analysis — 29 CFR 1926 OSHA citations | /contract-review — AIA A201 article-level risk | /toolbox-talk — weekly safety meeting format | /daily-log — construction daily report narrative | /estimate — returns JSON CSI line item array | /notice-draft — formal AIA contract notice letter | /prequalification — sub scoring narrative | /punch-list — closeout punch item analysis

RS Means Methodology: Phase 1 (current): AI generates estimates using RS Means methodology + City Cost Index factors | Charlotte NC CCI: Material 0.87 | Labor 0.72 | Total 0.80 | CSI MasterFormat division structure | Phase 2: Gordian OEM data license after 20-50 paying subscribers (~\$15K-40K/yr)

AI Cost Model: claude-sonnet-4: ~\$3/MTok input | ~\$15/MTok output | Typical AI call: ~1,000-2,000 tokens total | Cost per call: ~\$0.01-0.05 | At 100 subscribers using 50 AI calls/mo each: ~\$50-250/mo in AI costs | Well within margin at \$149/mo entry price

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Procore Export

POST /api/sync/procore-export/:projectId

Exports: submittals, RFIs, daily logs, change orders

Maps Pocket PM fields to Procore API schema

Owner can import — GC needs no Procore account

JSON package output — no active subscription required

ACC / Autodesk Export

POST /api/sync/acc-export/:projectId

RFIs mapped to ACC RFQ format

Ball-in-court field mapped to ACC assignee

Drawing sheet numbers to ACC discipline codes

Submittals mapped to ACC submittal status schema

RS Means (Phase 2)

Phase 1 — now: AI methodology + CCI factors

Phase 2 — Gordian OEM data license

Trigger: 20-50 paying subscribers

Estimated cost: \$15K-\$40K/yr OEM license

Alternative: Craftsman Book Co. ~\$5K-\$15K/yr

Product Roadmap: RS Means Gordian OEM license (Phase 2) | AI conversation history persistence to ai_sessions collection | WebSocket real-time life safety deficiency notifications | File upload UI for drawings and daily log photos | Mobile-responsive sidebar layout | Multi-project dashboard summary view | PMBOK Project Intelligence module | K-12 Education document system | Overlay Layered Floor Plans module | Training Platform

Competitive Position: Between lightweight residential tools (\$49-99/mo) and enterprise Procore (\$375+/mo per user) | Entry price: \$149/mo per company (not per seat) | Target: Small and mid-size GCs competing for commercial and industrial projects in the \$1M-\$25M range | Tagline: Bid Bigger. Build Smarter. Win More.

Banking & Finance: Primary operating: Mercury (no fees, ~4-5% treasury yield, startup integrations) | Credit products: Capital One Spark at MRR stage | Revenue milestone targets: 20 subs = RS Means negotiation | 50 subs = Gordian OEM | 100 subs = enterprise sales hire

Service	Purpose	Plan	Cost/mo
VPS (Hetzner)	PocketBase + Caddy + Express	CX21 — 2 vCPU / 4 GB RAM	~\$6
Domain	pocketpm.fyi DNS A records	Annual / 12	~\$1
SSL Certificate	Let's Encrypt via Certbot	Free — auto-renews 90 days	\$0
Anthropic API	Claude Sonnet 4 — all AI features	Pay per use	~\$10-50
RS Means Data	Phase 1: AI methodology Phase 2: Gordian OEM	OEM after 50 subscribers	\$0 now
TOTAL	At 100 subs x \$149/mo = \$14,900 MRR	Infra < 0.5% of revenue	~\$17-57

Key Design Decisions

- Single-file frontend:** No build step, no bundler, no node_modules in browser. The entire app ships as one HTML file. Deployments are a single scp command. Zero dependency rot. Python validation runs after every build.
- Static panel architecture:** No renderShell() — all 27 panels exist as static HTML, nav() shows/hides via display:none/block. No DOM rebuilding on navigation. Survives the v7 merge mistake that broke v8.
- API key security:** The Anthropic key lives exclusively in Railway's environment variables. Never sent to the browser, never in the HTML file, never in PocketBase. Browser only holds a short-lived JWT.
- Three-tier persistence:** Mock mode (localStorage) -> PocketBase direct -> Railway+PocketBase. The app is always demonstrable — no backend required for POC. Same codebase, same UI, different data source.
- Data ownership:** All project data lives in SQLite on your VPS. You can tar the entire database in one command. No vendor lock-in, no per-seat storage fees, no data hostage situations.
- Procore/ACC independence:** Export layers generate platform-compatible packages on demand. Owner can import into Procore without requiring the GC to pay for Procore. Core competitive differentiator vs enterprise tools.